Sign compression for Muon: SignMuon, MuonSign, and the Limits of Error Feedback

Anonymous submission

Abstract

SignMuon compresses the Muon update to one bit per parameter by taking its elementwise sign, providing the most direct way to run a matrix-aware optimizer under an extremely low communication budget. It outperforms SignSGD in practice, yet it can ascend even on a linear function. Signing the gradient *before* the Linear Minimization Oracle (LMO), rather than after, does not repair this: we construct a small explicit instance on which sign-before (MuonUSign) and sign-on-both-sides (MuonSign) ascend as well, so no placement of the sign around the oracle descends in general. Error feedback, the standard remedy for a biased compressor, does not rescue SignMuon: when applied to Muon's output, error feedback can fail for every smoothness constant, step size and momentum. Applied to the gradient, error feedback does work, and EF21-MuonUSign and EF21-MuonSign attain the standard $\mathcal{O}(T^{-1/2})$ rate for the squared gradient norm on smooth nonconvex problems, the latter at one bit in each direction. Experiments then reverse the ordering: across centralized CIFAR-10, federated CIFAR-10 and the nanoGPT speedrun, the strongest compressed method is consistently sign-*after*-the-LMO, precisely the placement we prove divergent, with the provably convergent variants trailing it. Compressing after the LMO, a heuristic, matters more at these scales than the guarantee does.

1 Introduction

Training a deep network across clients consumes bandwidth as well as computation: every round each client transmits a full update, and under a limited link those updates dominate the wall-clock cost. Compression is the standard remedy (Bernstein et al. 2018, 2019; Beznosikov et al. 2023), and SignSGD is its extreme point: one bit per coordinate, at slight cost in accuracy. SignSGD, however, flattens each weight matrix into a vector, discarding structure that other optimizers exploit. Muon exploits precisely that structure, orthonormalizing the momentum matrix before stepping, and in several settings surpasses tuned adaptive methods (Jordan et al. 2024b; Bernstein and Newhouse 2024; Shah et al. 2025). What it transmits, however, is a *dense* matrix at full precision: thirty-two bits per parameter where the budget allows one. Matrix geometry and a one-bit budget are therefore difficult to obtain together.

We study the natural ways to combine sign compression with the Muon LMO at one bit per parameter: **SignMuon**, which signs *after* the LMO, $\mathrm{sign}(\mathrm{LMO}(\cdot))$; **MuonUSign**, which signs *before*, $\mathrm{LMO}(\mathrm{sign}(\cdot))$; and **MuonSign**, which signs on *both* sides and, like SignMuon, emits a ± 1-valued step, so that uplink and downlink alike cost one bit. All three build Muon's matrix-aware geometry into the step without preserving it intact, and they are not interchangeable: on federated CIFAR-10 (Table 2) they span 2.8 accuracy points, in the order after, before, both sides, and only sign-after matches full-precision Muon.

None of them, however, descends in general. We prove that each can *ascend* on a linear objective, the simplest smooth function there is, at every step size and every momentum: SignMuon on a 4×4 gradient (Theorem 1), MuonUSign and MuonSign on a single 5×5 one (Theorems 2–3), both shapes minimal. The standard repair for a biased compressor is error feedback, and for sign-after it is unavailable: applied to the oracle's *output* it fails for every triple (L, η, μ): there is then an L-smooth objective on which the method diverges (Theorem 4), so no step-size rule built from the smoothness and momentum constants can save it.

What error feedback does repair is the placement that compresses the gradient instead. Adapting EF21-Muon (Gruntkowska et al. 2025) to sign compression gives **EF21-MuonUSign**, which drives $\min_{t \le T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2$ to zero at the standard $\mathcal{O}(T^{-1/2})$ rate for smooth nonconvex problems, at a one-bit uplink, and **EF21-MuonSign**, which adds a second error-feedback loop on the downlink so that both channels carry one bit for little further cost. Both descend on the counterexamples above. To our knowledge the sign-before and both-sides placements are new, as are the two error-feedback methods; SignMuon itself was proposed concurrently by Mishra, Trivedi, and Kumar (2026), whose guarantee is proved for the gradient-sign oracle and not for SignMuon (Appendix A.7).

We evaluate all six against Muon, SignSGD, SGD and Adam on centralized CIFAR-10 with a ResNet-18, federated CIFAR-10 at $N = 11$ clients, the nanoGPT speedrun, and a synthetic convex quadratic on which the quantity the counterexamples attack can be measured directly. Theory and experiment then disagree. On random quadratics the gradient/step alignment that Theorems 1–3 drive negative stays positive for all three placements, so the constructions describe a worst case that ordinary data does not produce; and on all three networks the best compressed method is sign-after, exactly the placement we prove unrepairable, the two provably convergent variants trailing it by several seed spreads. At these scales, compressing only after the oracle outweighs the guarantee. A tuned five-seed federated comparison establishes that ordering. Proofs and every result cited but not stated here are in the appendix, which continues this numbering.

2 Related Work

Sign compression. The sign is the most widely used compressor in this literature (Alistarh et al. 2017; Horváth et al. 2023; Beznosikov et al. 2023): one bit per coordinate, no index set or scale sent beside it, and a majority vote of signs that is again a sign, so both directions stay at one bit (Bernstein et al. 2019). Its theory is correspondingly well developed. SignSGD (Bernstein et al. 2018) needs growing batches to converge; its bias is otherwise repaired by error feedback (Seide et al. 2014; Karimireddy et al. 2019), sharpened into EF21 (Richtárik, Sokolov, and Fatkhullin 2021), by momentum (Cutkosky and Mehta 2020; Sun et al. 2023), or by randomizing the sign operator (Chen et al. 2020; Safaryan and Richtárik 2021; Jin et al. 2025).

LMO optimizers. Muon (Jordan et al. 2024b) is the spectral-norm instance of a norm-constrained LMO step (Bernstein and Newhouse 2024; Pethick et al. 2025), analysed by Li and Hong (2025) and Kovalev (2025), then generalized to arbitrary layer norms by Gluon (Riabinin et al. 2025).

Compressed and federated Muon. Gruntkowska et al. (2025) give the error-feedback framework for bidirectionally compressed Muon/Gluon that EF21-MuonUSign and EF21-MuonSign instantiate. Around it: Qian et al. (2026) compress Gluon with SARAH-type variance reduction; Takezawa et al. (2026) and Zhang and Gao (2025) study federated LMO steps without compression; Ahn et al. (2025) make the orthonormalization low-rank with error feedback, Gupta et al. (2025) quantize Muon's optimizer states, Thérien et al. (2025) quantize the delta of DiLoCo's Muon inner loop (Douillard et al. 2023) to two bits at next to no loss. Concurrently, Mishra, Trivedi, and Kumar (2026) proposed plain SignMuon with an $\mathcal{O}(1/\sqrt{T})$ guarantee, which Appendix A.7 shows to be a guarantee for a different method: the rate is proved only for the gradient-sign oracle, which is SignSGD's update and invokes no polar factor, while the one bound covering the polar-sign update carries an unestimated residual that exceeds the bounded quantity exactly where the expected step ascends, as it does throughout the instance of Theorem 1. Two further methods relate sign descent to Muon differently. Bolatov et al. (2026) *alternate* spectral and sign steps rather than composing them; Kravatskiy et al. (2025) mix the geometries *inside* the oracle, so S-Muon stays an LMO for an explicit norm, with the attendant theory. Our placements put the sign *around* the oracle, where it is a compressor and not a geometry, so the counterexamples below bear on neither (Appendix A.8).

3 Problem Statement

We consider the stochastic optimization problem

$$\min_{\mathbf{X}\in\mathcal{X}}\{f(\mathbf{X}) := \mathbb{E}_{\xi\sim\mathcal{D}}[f(\mathbf{X};\xi)]\}, \tag{1}$$

where $\mathcal{X}$ is the parameter space (e.g., $\mathbb{R}^d$ or $\mathbb{R}^{m\times n}$) and $f(\cdot\,;\xi):\mathcal{X}\to\mathbb{R}$ is continuously differentiable and possibly non-convex: $f(\mathbf{X};\xi)$ is the loss of a model $\mathbf{X}$ at a data point $\xi\sim\mathcal{D}$. In the federated setting the data are split across clients, and the training problem becomes

$$\min_{\mathbf{X}\in\mathcal{X}}\left\{f(\mathbf{X}) := \tfrac{1}{N}\textstyle\sum_{j=1}^{N} f_j(\mathbf{X})\right\}, \tag{2}$$

where $N\geq 1$ is the number of clients and $f_j(\mathbf{X}) = \mathbb{E}_{\xi_j\sim\mathcal{D}_j}[f_j(\mathbf{X};\xi_j)]$ is the loss on the data $\mathcal{D}_j$ held by client $j\in[N] := \{1,\ldots,N\}$.

The model is a tuple of layers $\mathbf{X} = [\mathbf{X}_1,\ldots,\mathbf{X}_p]$, $\mathbf{X}_i\in\mathbb{R}^{m_i\times n_i}$ with $m_i,n_i\geq 1$, with ∇_i the gradient block of layer i; iterates carry a time index, $\mathbf{X}_t$, and $\mathbf{X}_{t,i}$ is layer i of iterate t. This is Gluon's parameter space (Riabinin et al. 2025) under a single geometry: each layer carries the spectral norm $\|\cdot\|_{2\to2}$, whose dual is the nuclear norm $\|\cdot\|_*$, and we define $\|\nabla f(\mathbf{X})\|_*^2 := \sum_i\|\nabla_i f(\mathbf{X})\|_*^2$; a single matrix parameter is the case $p=1$. A vector parameter is a block of width one, on which the spectral norm is Euclidean and the oracle only normalizes, so all three methods of Section 4 coincide with SignSGD there and the divergence results below need $\min(m_i,n_i)\geq 2$ (Appendix A.2).

Two assumptions run through the paper.

Assumption 1 (Lower boundedness) *$f(\mathbf{X})\geq f^*$ for all $\mathbf{X}$; where explicitly invoked, each local $f_j\geq f_j^*$ as well.*

Assumption 2 (Layer-wise smoothness) *For every layer i and all $\mathbf{X},\mathbf{Y}$,*

$$\|\nabla_i g(\mathbf{X})-\nabla_i g(\mathbf{Y})\|_*\leq L_i^g\,\|\mathbf{X}_i-\mathbf{Y}_i\|_{2\to2} \tag{3}$$

for $g=f$ (constant L_i) and $g=f_j$ (constant $L_{i,j}$). A weaker layer-wise (L^0,L^1) form, stated in Appendix A.1, is used only in Corollary 2 (Appendix A.10). Both forms are quantified over arbitrary $\mathbf{X},\mathbf{Y}$, as in Gruntkowska et al. (2025); Appendix A.1 records what that strength implies.

A Linear Minimization Oracle (LMO) minimizes the first-order model of the objective over the unit ball of a norm, returning the steepest-descent direction in that norm (Appendix A.1). Muon is the oracle of the spectral norm fixed above: writing $\mathbf{M}=\mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$ for the rank-r singular value decomposition of $\mathbf{M}\in\mathbb{R}^{m\times n}$, the LMO direction is

$$A(\mathbf{M}) = -\mathbf{U}\mathbf{V}^\top\in\mathbb{R}^{m\times n},\qquad \mathrm{polar}(\mathbf{M}) := -A(\mathbf{M}), \tag{4}$$

an orthogonalization, unscaled. Muon's reference implementation rescales it by $\sqrt{\max(1,m/n)}$ (Jordan et al. 2024b) and the RMS$\to$RMS reading of the layer norm by $\sqrt{m/n}$: positive per-layer constants, invisible to $\mathrm{sign}(\cdot)$, which we carry in the step size (7) and not in the geometry (Appendix A.1). In practice the oracle is applied to a momentum estimate rather than to a gradient; (5) below gives the full iteration. What Muon must transmit is that dense direction, precisely what a bandwidth-limited link cannot accommodate; our object is therefore its matrix-aware geometry at a one-bit budget.

4 Theory

From SignSGD to Sign A. SignSGD was introduced as a compressed SGD: a convergent method, of which only the sign of the update is transmitted. Nothing in that construction is specific to SGD, so one may apply it to any optimizer that emits a direction, expecting the compressed method to inherit what the direction contributed. LMO-based optimizers give grounds for that expectation, their theory being uniform in the geometry: one analysis covers every norm, which enters only through its oracle and a pair of norm-equivalence constants (Pethick et al. 2025; Kovalev 2025; Riabinin et al. 2025). Were signing a direction harmless, it would be harmless across that family, and the member one seeks to compress is Muon.

The second ground for the expectation is where it fails. The sign step is itself an LMO, steepest descent in ℓ_∞ (Bernstein and Newhouse 2024), so signing a Muon direction composes two oracles, each sound alone. The composition is an LMO for *no* norm. An oracle for a norm $\|\cdot\|$ returns $\mathbf{D} = -A(\mathbf{G})$ with $\langle \mathbf{G}, \mathbf{D} \rangle = \max_{\|\mathbf{Y}\| \le 1} \langle \mathbf{G}, \mathbf{Y} \rangle = \|\mathbf{G}\|_{\mathrm{dual}}$, nonnegative since $\mathbf{0}$ is feasible; Theorem 1 provides a gradient on which the composition drives that inner product strictly negative. The guarantee is forfeited in the composition, not in either factor.

The Sign A family. Let A be any optimizer built on LMO directions. A Sign A method initializes $\mathbf{M}_0 = \mathbf{0}$ and at each iteration $t \ge 1$ performs

$$\begin{aligned}
&\mathbf{M}_t = \mu \mathbf{M}_{t-1} + (1-\mu)\, \mathbf{G}_t, \\
&\tilde{\mathbf{M}}_t = \mathbf{M}_t \ \text{ or } \ (1-\mu)\mathbf{G}_t + \mu \mathbf{M}_t \ \text{ (Nesterov)}, \\
&\mathbf{D}_t = -A(\tilde{\mathbf{M}}_t), \qquad \mathbf{s}_t = \operatorname{sign}(\mathbf{D}_t) \in \{\pm 1\}^{m \times n}, \\
&\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t,
\end{aligned} \tag{5}$$

where $\mathbf{X}_t \in \mathbb{R}^{m \times n}$ is the parameter matrix, $\mathbf{G}_t = \nabla f(\mathbf{X}_{t-1}; \xi_t)$ the stochastic gradient at the current iterate, $\mu \in [0, 1)$ the momentum coefficient and η_t the learning rate; $A(\cdot)$ is the LMO, and $\operatorname{sign}(\cdot)$ acts element-wise on the structured direction $\mathbf{D}_t$, with $\operatorname{sign}(0)$ resolved to an independent random ± 1 so that the transmitted alphabet stays binary. Momentum is in exponential-moving-average form throughout; the heavy-ball form differs by the positive factor $1/(1-\mu)$, which $\operatorname{sign}(\cdot)$ and the LMO both absorb, so no result below distinguishes them.

Three placements of the sign. Of the three, only SignMuon is a Sign A method. All three keep the momentum of (5) and the update $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \, \mathbf{s}_t$, and differ only in where the sign acts on $\tilde{\mathbf{M}}_t$:

$$\mathbf{s}_t = \begin{cases} \operatorname{sign}\big(\operatorname{polar}(\tilde{\mathbf{M}}_t)\big) & \text{(SignMuon: after)}, \\ \operatorname{polar}\big(\operatorname{sign}(\tilde{\mathbf{M}}_t)\big) & \text{(MuonUSign: before)}, \\ \operatorname{sign}\big(\operatorname{polar}(\operatorname{sign}(\tilde{\mathbf{M}}_t))\big) & \text{(MuonSign: both)}. \end{cases} \tag{6}$$

The three are Algorithms A2, A4 and A5 (Appendix A.18), all transmitting one bit per matrix parameter on the uplink. On the downlink SignMuon and MuonSign distribute a ± 1-valued object and so cost one bit each way, whereas MuonUSign's server-side $\operatorname{polar}(\cdot)$ is dense and goes at full precision.

4.1 Centralized Learning

Centrally, SignMuon is applied to the matrix-valued parameters ($n_{dim} \ge 2$), while vector parameters (biases, BatchNorm) and the final classification layer are trained with AdamW, the design of Jordan et al. (2024b), whose LMO branch is what Li and Hong (2025) and Riabinin et al. (2025) analyse. We approximate $\operatorname{polar}(\cdot)$ by a 5th-order Newton–Schulz iteration rather than a full SVD (Algorithm A1). One further implementation choice has consequences for every experiment below.

Per-layer step sizes: the unit-gain heuristic. Write $\mathbf{P}_\ell$ for the matrix a method applies to layer ℓ, of shape $m \times n$ (output $\times$ input dimension). It belongs to one of two families, each of *exactly* known Frobenius norm: $\mathbf{P}_\ell = \mathbf{U}\mathbf{V}^\top$ with $\|\mathbf{P}_\ell\|_F = \sqrt{r}$, $r = \min(m, n)$, for the LMO-terminated methods, and $\mathbf{P}_\ell = \mathbf{s}_\ell \in \{\pm 1\}^{m \times n}$ with $\|\mathbf{P}_\ell\|_F = \sqrt{mn}$ for the SIGN-terminated ones. The two scale differently with layer shape, so no single global η is appropriate for both families, or across layers within one. We fix the shape dependence a priori and tune only a shape-free base rate η_0, giving layer ℓ the step size $\eta_\ell = \eta_0 \lambda_\ell$ with

$$\lambda_\ell = \frac{\sqrt{m}}{\|\mathbf{P}_\ell\|_F} \quad \Longrightarrow \quad \begin{cases} \lambda_\ell^{\text{LMO}} = \sqrt{\max(1, m/n)}, \\ \lambda_\ell^{\text{SIGN}} = 1/\sqrt{n}. \end{cases} \tag{7}$$

The criterion is that every layer's update have the same root-mean-square input–output gain, the average-case form of the spectral scaling condition (Yang, Simon, and Bernstein 2023; Large et al. 2024), so that η_0 is the per-step RMS gain. Its LMO branch is not new: it reproduces the aspect-ratio factor that the reference Muon implementation already applies (Jordan et al. 2024b). That agreement is the external check we rely on, and it is what licenses applying the same criterion to the SIGN family, for which no such convention exists. We treat (7) as a heuristic and apply it uniformly. Appendix A.17 gives the derivation and the measurement selecting the exponent $\frac{1}{2}$ in $\lambda_\ell^{\text{SIGN}} = n^{-a}$ over μP's 1 (Yang et al. 2021).

4.2 Divergence of the Three Sign Placements

The descent property of the Muon LMO direction is lost under sign compression, before or after the oracle: in each of the three placements the compressed step can become an *ascent* direction on a smooth objective. We refute the descent property on *linear* objectives, the simplest smooth functions:

$$f(\mathbf{X}) = \langle \mathbf{G}, \mathbf{X} \rangle = \operatorname{Tr}(\mathbf{G}^\top \mathbf{X}), \qquad \nabla f(\mathbf{X}) \equiv \mathbf{G}. \tag{8}$$

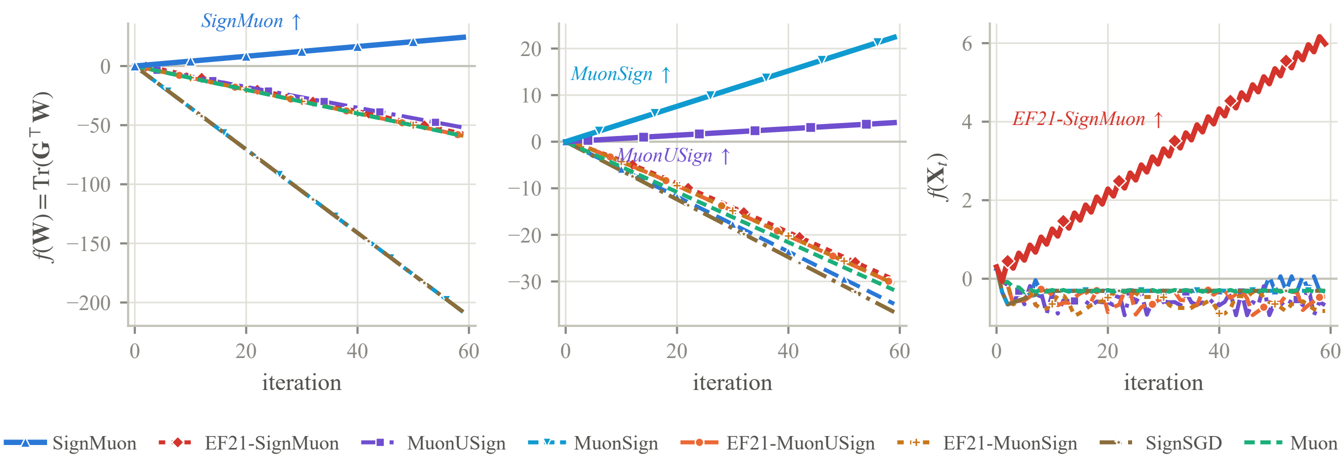

Figure 1: All eight methods on the three counterexample instances; the ascending method is drawn heavy and named in each panel. *Left:* SignMuon, 4×4 (Theorem 1). *Centre:* MuonUSign and MuonSign on their shared 5×5 instance (Theorems 2–3); both panels plot (8). *Right:* EF21-SignMuon, 2×2 (Theorem 4), normalized units at $\eta = 1$. Trajectories are momentum-free without loss (Proposition 1; Lemma 4 for EF21-SignMuon).

Gradient descent and full-precision Muon drive $f \to -\infty$ here, so a rule that instead drives $f \to +\infty$ is unambiguously ascending (Remark 1 in Appendix A.9 restores Assumption 1 without moving any trajectory below). On (8) the step is the constant matrix $\mathbf{s}(\mathbf{G})$ whatever the momentum, so momentum affects neither convergence nor divergence (Proposition 1, Appendix A.3) and

$$f(\mathbf{X}_t) - f(\mathbf{X}_{t-1}) = -\eta_t \left\langle \mathbf{G},\ \mathbf{s}(\mathbf{G}) \right\rangle \tag{9}$$

for every $\mu \in [0, 1)$ and either momentum rule. A single $\mathbf{G}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ therefore makes f increase at every step, and such a $\mathbf{G}$ exists for each placement at small size.

Theorems 1–3 (summary). *There exists $\mathbf{G} \in \mathbb{R}^{4\times4}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ for SignMuon, and $\mathbf{G} \in \mathbb{R}^{5\times5}$ with $\langle \mathbf{G}, \mathbf{s}(\mathbf{G}) \rangle < 0$ for MuonUSign and for MuonSign simultaneously. On the corresponding objective* (8) *each method ascends at every iteration, for every $\eta_t > 0$, every $\mu \in [0, 1)$ and either momentum rule, and $f(\mathbf{X}_t) \to +\infty$ whenever $\sum_t \eta_t = \infty$.*

Both instances are explicit and both inner products exact rationals; Appendices A.4–A.6 state the theorems, prove them, and bound the two shapes from below. Figure 1 (left, centre) demonstrates the ascents. Negative results of this kind exist for uncompressed Muon as well: Parshakova et al. (2026) show that it need not converge on convex Lipschitz problems. Ours are due to the compressor rather than the geometry, and hold on a smooth objective.

4.3 Centralized Error Feedback: EF21-SignMuon

Theorems 1–3 rule out all three placements of the sign around the oracle. The classical remedy for a biased compressor is *error feedback*: keep what the compressor discarded and fold it into the next message (Seide et al. 2014; Karimireddy et al. 2019). We work throughout in its EF21 form (Richtárik, Sokolov, and Fatkhullin 2021), which stores an estimator and compresses the *difference* to it, and so needs no bounded-gradient assumption. Its most direct use for SignMuon keeps the geometry, the sign *after* the LMO, and applies error feedback to the oracle's output. The resulting method, **EF21-SignMuon** (Algorithm A3), tracks an estimate $\mathbf{d}_t^{\mathrm{est}}$ of the polar factor $\mathbf{D}_t = \mathrm{polar}(\tilde{\mathbf{M}}_t)$, updated by a scaled sign of the residual,

$$\begin{aligned} \mathbf{d}_t^{\mathrm{est}} &= \mathbf{d}_{t-1}^{\mathrm{est}} + \alpha_t \,\mathrm{sign}\big(\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}\big), \\ \alpha_t &= \mathrm{mean}\big|\mathbf{D}_t - \mathbf{d}_{t-1}^{\mathrm{est}}\big|, \end{aligned} \tag{10}$$

and steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t\, \mathbf{d}_t^{\mathrm{est}}$.

Error feedback does not repair this placement: for *every* step size and momentum setting there is a smooth objective on which EF21-SignMuon diverges.

Theorem 4 (Divergence of EF21-SignMuon) *For every $L > 0$, step size $\eta > 0$, momentum coefficient $\mu \in [0, 1)$, and either momentum variant, there is an L-smooth (Assumption 2), bounded-below (Assumption 1) function $f : \mathbb{R}^{2\times2} \to \mathbb{R}$ on which EF21-SignMuon started at $\mathbf{X}_0 = \mathbf{0}$ diverges: for an explicit constant $c = c(f) > 0$,*

$$f(\mathbf{X}_{t+2}) - f(\mathbf{X}_t) = c\, L\eta^2 > 0 \qquad (t \geq 3), \tag{11}$$

so $f(\mathbf{X}_t) \to +\infty$. In particular, no step-size rule $\eta = \eta(L, \mu)$ using only the smoothness and momentum constants can make the method convergent.

The construction (Appendix A.9) turns the shared magnitude against the method. Its 2×2 LMO target has a large off-diagonal that reverses sign every step, holding the residual, and with it α_t, at $\Theta(1)$, and a small constant on the diagonal, which that magnitude overshoots at every step; the diagonal estimate settles into a period-two cycle whose average has the *wrong* sign, and the coordinate it drives diverges. The objective is not convex, unlike the linear ones of Theorems 1–3: sustaining the cycle requires a target sequence that never settles, which here a bounded periodic field supplies.

Dimension is not implicated, 2×2 being where the scaled sign's worst-case contraction $1/d$ is most favourable, nor is momentum: Figure A2 (Appendix A.9) records the same rate for every μ and both variants.

4.4 Centralized Error Feedback: EF21-MuonUSign and EF21-MuonSign

The shared magnitude α_t is not itself the defect: the two methods below couple all coordinates through the same scalar and converge. What fails is the target. Error feedback needs an estimator whose target varies with the step size, and the polar factor $\operatorname{polar}(\tilde{\mathbf{M}}_t)$ is not one: it is not Lipschitz in its argument, so it can move by $\Theta(1)$ between consecutive rounds however small η is. Compressing the *gradient* in its place restores that dependence, and the mechanism is then EF21-Muon (Gruntkowska et al. 2025), the framework within which we apply sign compression.

That framework compresses the two directions of the link separately, through a pair $(\mathcal{C}^{\uparrow}, \mathcal{C}^{\downarrow})$ of contractive compressors. In both directions we take the scaled sign $\mathcal{C}(\mathbf{Y}) = \operatorname{mean}|\mathbf{Y}| \operatorname{sign}(\mathbf{Y})$: one bit per parameter, plus one scalar per layer. **EF21-MuonUSign** takes the pair $(\mathcal{C}, I)$, a compressed uplink and a full-precision model back, the appropriate allocation when only the uplink is constrained. It maintains a gradient estimator $\mathbf{g}_t^{\text{est}}$, updated from the residual $\Delta_t = \tilde{\mathbf{M}}_t - \mathbf{g}_{t-1}^{\text{est}}$:

$$\alpha_t = \operatorname{mean}(|\Delta_t|), \quad \mathbf{g}_t^{\text{est}} = \mathbf{g}_{t-1}^{\text{est}} + \alpha_t \operatorname{sign}(\Delta_t), \tag{12}$$

and takes the step $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t$ with $\mathbf{D}_t = -A(\mathbf{g}_t^{\text{est}}) \approx \mathbf{U}_t \mathbf{V}_t^{\top}$, the polar factor of the estimator $\mathbf{g}_t^{\text{est}}$ rather than of ∇f. The sign in (12) acts on the internal compression residual alone, so the uplink stays at one bit per parameter (Algorithm A6).

EF21-MuonSign takes the pair $(\mathcal{C}, \mathcal{C})$: one bit in each direction. Its downlink compressor is a second error-feedback loop, on the *model* rather than the gradient, and it splits the method into two iterates:

$$\begin{aligned} \mathbf{X}_t &= \mathbf{X}_{t-1} - \eta_t \mathbf{D}_t && \text{(server model)}, \\ \mathbf{W}_t &= \mathbf{W}_{t-1} + \alpha_t^{\downarrow} \operatorname{sign}(\mathbf{X}_t - \mathbf{W}_{t-1}) && \text{(broadcast model)}, \end{aligned} \tag{13}$$

with $\mathbf{D}_t = -A(\mathbf{g}_t^{\text{est}})$ as before, and $\alpha_t^{\downarrow} = \operatorname{mean}|\mathbf{X}_t - \mathbf{W}_{t-1}|$. The server model $\mathbf{X}$ takes the exact LMO step and never leaves the server; what crosses the downlink is the one-bit update of $\mathbf{W}$, the only model the rest of the method observes, since gradients, momentum and the uplink residual are all computed there, whereas the guarantee bounds the gradient at $\mathbf{X}$. The rate is preserved under the second loop at a step size smaller by a factor of $\sqrt{r}$ in the layer rank, a penalty the analysis cannot avoid because the scaled sign contracts in the Euclidean norm and not in the layer norm the spectral geometry requires (Remark 5, Appendix A.10); Section 5.3 measures what it costs in practice.

Both methods descend where the placements they repair ascend: Figure 1 (centre) has them on the 5×5 instance of Theorems 2–3. Appendix A.10 shows them to be *exact instances* of the EF21-Muon framework, the one substantive check being that the scaled sign is a contractive compressor (Lemma 5), so they inherit its guarantees (Theorem 5). Under Assumptions 1–2 both reach $\min_{t \le T} \mathbb{E}\|\nabla f(\mathbf{X}_t)\|_*^2 = \mathcal{O}(T^{-1/2})$, the standard smooth nonconvex rate, which uncompressed Muon attains as well: one bit changes the constant and not the rate. Under (L^0, L^1)-smoothness EF21-MuonUSign reaches $\min_t \sum_i \mathbb{E}\|\nabla_i f(\mathbf{X}_t)\|_* = \mathcal{O}(T^{-1/4})$ at a constant step size (Corollaries 1–2). Memory is the other cost, one model-sized buffer per compressed channel: EF21-MuonUSign holds the gradient estimator on each client, and EF21-MuonSign holds that estimator and, on the server, the broadcast model $\mathbf{W}$. Of these, only the client-side buffer is a practical constraint, since the server is the better-provisioned side.

4.5 Federated Learning

In the federated setting (2) the placement decides where the oracle runs. With the sign *after* it, each client must orthogonalize its own momentum: it computes a stochastic gradient at the broadcast model, maintains a momentum buffer, applies the Muon LMO (Algorithm A1), then uploads the elementwise sign of the result. The server takes a majority vote, $\mathbf{s}_t^{\text{agg}} = \operatorname{sign}\big(\sum_j \mathbf{s}_t^{(j)}\big)$, and steps $\mathbf{X}_t = \mathbf{X}_{t-1} - \eta_t \mathbf{s}_t^{\text{agg}}$, as SignSGD does (Bernstein et al. 2019). With the sign *before*, the client uploads $\operatorname{sign}(\tilde{\mathbf{M}}_t^{(j)})$; the server votes, then applies a single LMO to the outcome. The vote rather than the average is what keeps the oracle's argument a sign matrix, so that the server-side method is exactly the MuonUSign of (6); the error-feedback methods below instead average, as their framework prescribes. The direction that returns is dense, so MuonUSign broadcasts the model at full precision, whereas MuonSign signs that direction once more and broadcasts $\operatorname{sign}(\mathbf{D}_t)$. SignMuon and MuonSign therefore send a ± 1-valued object down as well as up, the vote itself in the first case, free of ties at an odd client count, so both directions cost one bit; and since every client applies the same update, copies started from a common $\mathbf{X}_0$ never drift.

Error feedback changes the uplink message, not where the oracle runs: each client sign-compresses the residual between its estimator and the quantity it would otherwise have sent, the polar factor for EF21-SignMuon and the momentum for the other two, and sends $(\mathbf{s}_t^{(j)}, \alpha_t^{(j)})$; the server averages these into a global estimator. The extra per-layer scalar leaves the uplink at $\approx$1 bit per parameter, but that estimator is a scaled average of signs and so is dense, and the vote argument lapses: the full model must be broadcast unless a second error-feedback loop compresses it, as EF21-MuonSign does. Error feedback meets the same obstruction for SignSGD, where it is analysed for a single worker only (Karimireddy et al. 2019) and its distributed forms compress the return path with a second loop of their own (Tang et al. 2019). The complete procedures are Algorithms A8–A9.

5 Experiments

Descent in practice. On a deterministic convex quadratic (Appendix A.12) the first-order term of the descent lemma can be measured directly, through the alignment $\rho_t = \langle \nabla f, \mathbf{D}_t \rangle / (\|\nabla f\|_F \|\mathbf{D}_t\|_F)$ between the gradient and the di-

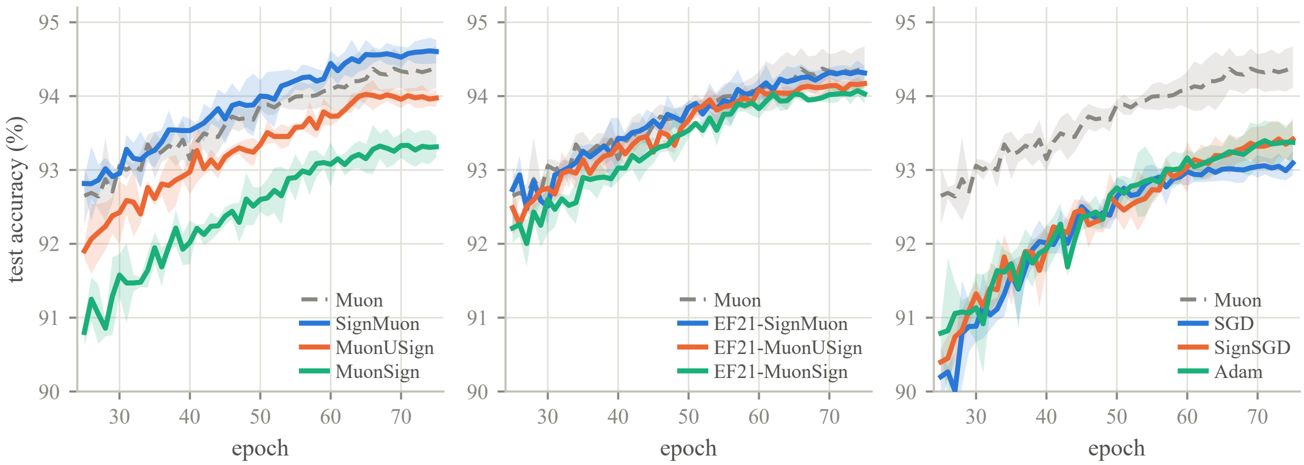

Figure 2: Centralized ResNet-18 on CIFAR-10: test accuracy from epoch 25 at each method's selected η_0 (Table 1). Panels group the sign placements, the EF21 variants and the baselines; Muon is the gray dashed reference, bands ±1 s.d. over three seeds.

Method	η_0	**Test acc (%)**	**Ep. to 90%**
SignMuon	0.02	94.60 ± 0.15	7.7
Muon	0.1	94.35 ± 0.27	7.7
EF21-SignMuon	0.02	94.31 ± 0.11	9.0
EF21-MuonUSign	0.05	94.14 ± 0.07	10.3
EF21-MuonSign	0.005	94.04 ± 0.10	11.0
MuonUSign	0.02	93.98 ± 0.12	10.3
Adam	0.001	93.37 ± 0.27	20.0
SignSGD	0.002	93.37 ± 0.26	19.0
MuonSign	0.1	93.31 ± 0.21	17.7
SGD	0.02	93.04 ± 0.14	21.3

Table 1: Centralized ResNet-18 / CIFAR-10, 75 epochs. Test accuracy is the mean $\pm$ s.d. over three seeds of the last five epochs; "Ep. to 90%" is the mean epoch at which it first reaches 90%.

rection actually taken. The direction descends when $\rho_t > 0$, and Theorems 1–3 provide instances on which $\rho_t < 0$; on random instances this does not occur, all three placements keeping ρ_t bounded away from zero at every step. The counterexamples describe a worst case rather than a typical one, which is what permits the network results below to run contrary to them.

5.1 Centralized Learning

We compare all six sign-based methods against Muon, SignSGD, SGD and Adam on CIFAR-10 (Krizhevsky 2009) with a ResNet-18 adapted to low-resolution images, over 75 epochs with η_0 cosine-annealed to zero. The only tuned hyperparameter is η_0, selected per method on a held-out split and read identically across methods through the unit-gain rule (7). Numbers average three seeds; differences below the seed spread are not claimed (Appendix A.11).

SignMuon ranks first, but by less than one seed spread over Muon: what Table 1 supports is that it *matches* Muon at one bit per parameter, not that it exceeds it. The separation that does resolve is the 1.2 points over SignSGD, which spends the same budget without the LMO; the geometry accounts for it, not the compressor. Error feedback is not free here, EF21-SignMuon lying 0.29 points below SignMuon against standard deviations of 0.11 and 0.15. Both families order the placements after, before, both sides, the last no better than SignSGD. The threshold column separates the methods more sharply than accuracy does: 7.7 epochs to 90% for SignMuon against 19.0–21.3 for SGD, SignSGD and Adam.

5.2 Federated Learning

We compare the same ten methods on CNN2 (two convolutional blocks with BatchNorm and an MLP classifier), with CIFAR-10 split homogeneously across $N = 11$ clients and no local steps; N is odd, so the majority vote cannot tie. Rates are tuned per method at the reporting horizon on a split held out *before* the client partition; two schemes cover all six, worker- and server-side LMO (Algorithms A8–A9, Table A10).

Federation separates the methods far more. At one bit per parameter in *both* directions SignMuon reaches 85.72% against Muon's 85.98%, about one seed standard deviation apart, and exceeds SignSGD by 4.3 points, far more than centrally. Five seeds resolve the three placements: they span 2.8 points in the order after, before, both sides, all three clearing SignSGD, the last by 1.5 points where centrally it was level. The second error-feedback loop is not the source of the cost: EF21-MuonSign, compressed in both directions, stands marginally ahead of the uplink-only EF21-MuonUSign.

Less favourably for the theory, the two methods carrying an unconditional guarantee trail SignMuon by several seed spreads, as they do centrally, so what the guarantee costs here is charged to the placement. EF21-MuonSign is scored at its server model $\mathbf{X}_t$, not the model its clients hold (13); the two

Optimizer	η_0	Up	Down	Rds. to 80%	Test acc (%)
Muon	0.1	32	32	260	85.98 ± 0.26
SignMuon	0.1	**1.09**	**1.09**	540	85.72 ± 0.24
EF21-SignMuon	0.02	**1.09**	32	640	84.71 ± 0.15
MuonUSign	0.05	**1.09**	32	500	84.56 ± 0.35
EF21-MuonSign	0.05	**1.09**	**1.09**	980	83.99 ± 0.28
EF21-MuonUSign	0.01	**1.09**	32	900	83.56 ± 0.41
MuonSign	0.02	**1.09**	**1.09**	860	82.94 ± 0.19
SGD	0.05	32	32	1240	81.69 ± 0.37
SignSGD	0.01	**1.09**	**1.09**	1140	81.44 ± 0.15
Adam	0.001	32	32	—	77.12 ± 1.04

Table 2: CIFAR-10 federated learning on CNN2, ordered by accuracy: $N = 11$ clients, homogeneous split, 2000 rounds at batch 192, momentum 0.9. Test accuracy is the mean of the final five evaluations over five seeds, $\pm$ one s.d. *across* seeds; Up and Down are bits per parameter per round each way (Appendix A.14). Every method but Adam reached 80% on all five seeds, Adam on none.

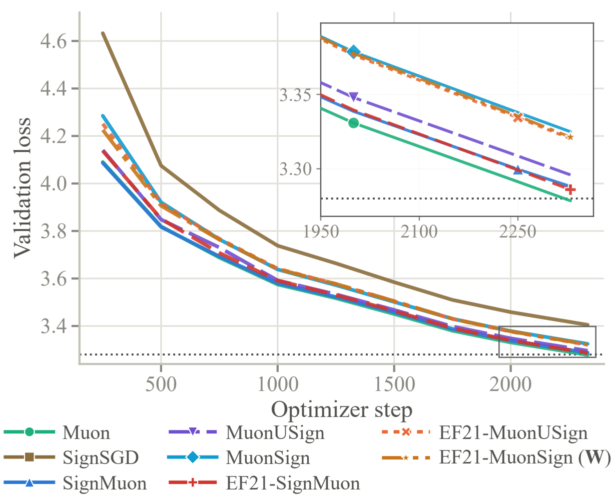

Figure 3: NanoGPT speedrun, $8\times$H100: validation loss against optimizer step, one run per method; dotted is the target 3.28, and EF21-MuonSign is drawn at its broadcast model $\mathbf{W}$. The inset magnifies the boxed tail, where SignSGD lies above the range shown.

track each other here, and Section 5.3 shows that they need not (Remark 4, Appendix A.10).

5.3 Language Modelling

We test where the matrices are large enough for the layer-rank dependence of Corollary 1 to take effect: the modded-nanoGPT speedrun (record #40), a 12-layer transformer with 768×3072 hidden matrices, trained on 611M FineWeb tokens on $8\times$H100 (Jordan et al. 2024a). Each method replaces the record's Muon on the hidden matrices and gates, leaving all else untouched; our Muon run is record #40's own update rule, re-implemented without its Triton kernels. Rates are fixed a priori by the unit-gain rule at one η_0 per *family*, so every contrast is matched-hyperparameter (Appendix A.16).

Method	Step	η_0	Val. loss	Steps to 3.35
Muon (record #40)	LMO	0.06	**3.2785**	$1.00\times$
EF21-SignMuon	LMO	0.06	3.2860	$1.02\times$
SignMuon	SIGN	0.03	3.2881	$1.02\times$
MuonUSign	LMO	0.06	3.2959	$1.05\times$
EF21-MuonUSign	LMO	0.06	3.3203	$1.13\times$
EF21-MuonSign ($\mathbf{W}$)	LMO	0.06	3.3213	$1.14\times$
MuonSign	SIGN	0.03	3.3249	$1.14\times$
SignSGD	SIGN	0.03	3.4049	–
EF21-MuonSign ($\mathbf{X}$)	LMO	0.06	5.5198	–

Table 3: NanoGPT after 2330 steps (611M tokens), one run per method. Record #40 reports 3.2780 ± 0.0009 over five seeds, so differences below $\sim$0.003 are noise; "Steps to 3.35" is relative to Muon. The last row is the server model $\mathbf{X}$ of the same run as the $\mathbf{W}$ row.

Three conclusions follow (Table 3, Figure 3). First, composing the sign with the LMO transfers to language modelling: all six such methods improve on SignSGD by at least 0.08 in validation loss, and the best two, EF21-SignMuon and SignMuon, lie within 0.01 of full-precision Muon at equal wall-clock. Second, on the sign-after placement error feedback is free: EF21-SignMuon and SignMuon differ by 0.002, below the $\sim$0.003 noise level, so the two are indistinguishable and jointly the strongest compressed methods here. The target both compress, $\mathrm{polar}(\tilde{\mathbf{M}}_t)$, is an orthogonal factor whose entries stay evenly spread in these runs, the regime in which the scaled sign loses least (Appendix A.16). Third, EF21-MuonSign's two models (13) separate at this width. The run has a single client, so its broadcast model $\mathbf{W}$ is the only point at which a gradient is ever evaluated, and $\mathbf{W}$ is indistinguishable from EF21-MuonUSign: the compressed downlink costs nothing where training occurs. The server model $\mathbf{X}$, the iterate the guarantee bounds, settles 2.2 nats above it, a persistent offset that Appendix A.16 localizes to one layer type, the zero-initialized output projection of each MLP block (`c_proj`). This is the mechanism of Remark 4 (Appendix A.10), not a tuning failure: only a spectrally contractive downlink compressor removes it.

The three settings therefore agree on the placement, and the placement they select is the one the theory excludes. Sign-after is what Theorem 1 makes diverge and error feedback repairs for no (L, η, μ) (Theorem 4), yet in all three the strongest compressed method is a sign-after one, ahead of every variant carrying an unconditional guarantee.

6 Conclusion

A sign step and a spectral step are each an LMO, sound alone; composed, they are an LMO for no norm, and the descent property each factor guarantees is lost in the composition. On small explicit instances, at every step size and momentum, every placement of the sign around the oracle can turn the update into an ascent direction on a linear objective: SignMuon after the LMO, MuonUSign before it, and MuonSign on both sides. Where error feedback is applied then decides whether it repairs this. On the oracle's *output* it does not: the polar factor can move by a constant however small the step size, so one shared magnitude cannot track it, and for every (L, η, μ) some L-smooth objective makes EF21-SignMuon diverge. On the *gradient* it does: EF21-MuonUSign attains the standard nonconvex rate at a one-bit uplink and EF21-MuonSign at one bit in each direction. Experiment ranks the methods in the opposite order, the strongest compressed method on all three architectures being a sign-after one, which carries no theoretical guarantee. We document that tension rather than resolve it: what the counterexamples preclude is a guarantee covering every problem in the class, not one holding under further conditions that ordinary problems satisfy. Identifying such conditions is the main open question, together with the convergence rate for the (L^0, L^1) setting under a compressed downlink and whether *any* one-bit compressor contracts in a layer norm.

References

Ahn, K.; Xu, B.; Abreu, N.; Fan, Y.; Magakyan, G.; Sharma, P.; Zhan, Z.; and Langford, J. 2025. Dion: Distributed Orthonormalized Updates. arXiv:2504.05295.

Alistarh, D.; Grubic, D.; Li, J.; Tomioka, R.; and Vojnovic, M. 2017. QSGD: Communication-Efficient SGD via Gradient Quantization and Encoding. In *Advances in Neural Information Processing Systems (NeurIPS)*, 1709–1720.

Bernstein, J.; and Newhouse, L. 2024. Old Optimizer, New Norm: An Anthology. arXiv:2409.20325.

Bernstein, J.; Wang, Y.-X.; Azizzadenesheli, K.; and Anandkumar, A. 2018. signSGD: Compressed Optimisation for Non-Convex Problems. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, 560–569. PMLR.

Bernstein, J.; Zhao, J.; Azizzadenesheli, K.; and Anandkumar, A. 2019. signSGD with Majority Vote is Communication Efficient and Fault Tolerant. In *International Conference on Learning Representations (ICLR)*. arXiv:1810.05291.

Beznosikov, A.; Horváth, S.; Richtárik, P.; and Safaryan, M. 2023. On Biased Compression for Distributed Learning. *Journal of Machine Learning Research*, 24(276): 1–50.

Bolatov, A.; Riabinin, A.; Kornilov, N.; Veprikov, A.; Horváth, S.; Takáč, M.; and Beznosikov, A. 2026. Lion-Muon: Alternating Spectral and Sign Descent for Efficient Training. arXiv:2605.19811.

Chen, X.; Chen, T.; Sun, H.; Wu, Z. S.; and Hong, M. 2020. Distributed Training with Heterogeneous Data: Bridging Median- and Mean-Based Algorithms. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:1906.01736.

Cutkosky, A.; and Mehta, H. 2020. Momentum Improves Normalized SGD. In *Proceedings of the 37th International Conference on Machine Learning*, volume 119 of *Proceedings of Machine Learning Research*, 2260–2268. PMLR.

Douillard, A.; Feng, Q.; Rusu, A. A.; Chhaparia, R.; Donchev, Y.; Kuncoro, A.; Ranzato, M.; Szlam, A.; and Shen, J. 2023. DiLoCo: Distributed Low-Communication Training of Language Models. arXiv:2311.08105.

Gruntkowska, K.; Gaponov, A.; Tovmasyan, Z.; and Richtárik, P. 2025. Error Feedback for Muon and Friends. arXiv:2510.00643.

Gupta, A.; Celente, R.; Shivanna, A.; Braithwaite, D. T.; Dexter, G.; Tang, S.; Udagawa, H.; Silva, D.; Ramanath, R.; and Keerthi, S. S. 2025. Effective Quantization of Muon Optimizer States. arXiv:2509.23106.

Horváth, S.; Kovalev, D.; Mishchenko, K.; Richtárik, P.; and Stich, S. 2023. Stochastic distributed learning with gradient quantization and double-variance reduction. *Optimization Methods and Software*, 38(1): 91–106.

Jin, R.; Liu, Y.; Huang, Y.; He, X.; Wu, T.; and Dai, H. 2025. Sign-Based Gradient Descent With Heterogeneous Data: Convergence and Byzantine Resilience. *IEEE Transactions on Neural Networks and Learning Systems*, 36(2): 3834–3846.

Jordan, K.; Bernstein, J.; Rappazzo, B.; @fernbear.bsky.social; Boza, V.; You, J.; Cesista, F.; Koszarsky, B.; and @Grad62304977. 2024a. modded-nanogpt: Speedrunning the NanoGPT baseline. https://github.com/KellerJordan/modded-nanogpt.

Jordan, K.; Jin, Y.; Boza, V.; You, J.; Cesista, F.; Newhouse, L.; and Bernstein, J. 2024b. Muon: An optimizer for hidden layers in neural networks. https://kellerjordan.github.io/posts/muon/.

Karimireddy, S. P.; Rebjock, Q.; Stich, S.; and Jaggi, M. 2019. Error Feedback Fixes SignSGD and other Gradient Compression Schemes. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, 3252–3261. PMLR.

Kovalev, D. 2025. Understanding Gradient Orthogonalization for Deep Learning via Non-Euclidean Trust-Region Optimization. arXiv:2503.12645.

Kravatskiy, A.; Kozyrev, I.; Kozlov, N.; Vinogradov, A.; Merkulov, D.; and Oseledets, I. 2025. Ky Fan Norms and Beyond: Dual Norms and Combinations for Matrix Optimization. arXiv:2512.09678.

Krizhevsky, A. 2009. Learning Multiple Layers of Features from Tiny Images. Technical report, University of Toronto. https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf.

Large, T.; Liu, Y.; Huh, M.; Bahng, H.; Isola, P.; and Bernstein, J. 2024. Scalable Optimization in the Modular Norm. In *Advances in Neural Information Processing Systems (NeurIPS)*. arXiv:2405.14813.

Li, J.; and Hong, M. 2025. A Note on the Convergence of Muon. arXiv:2502.02900.

Mishra, N.; Trivedi, K.; and Kumar, P. 2026. SignMuon: Communication-Efficient Distributed Muon Optimization. arXiv:2605.16311.

Parshakova, T.; Khaled, A.; Crawshaw, M.; Garrigos, G.; and Gower, R. M. 2026. Muon Does Not Converge on Convex Lipschitz Functions. arXiv:2605.08980.

Pethick, T.; Xie, W.; Antonakopoulos, K.; Zhu, Z.; Silveti-Falls, A.; and Cevher, V. 2025. Training Deep Learning Models with Norm-Constrained LMOs. In *Proceedings of the 42nd International Conference on Machine Learning*, volume 267 of *Proceedings of Machine Learning Research*, 49069–49104. PMLR.

Qian, X.; Gaponov, A.; Malinovsky, G.; and Richtárik, P. 2026. Communication-Efficient Gluon in Federated Learning. arXiv:2604.10689.

Riabinin, A.; Shulgin, E.; Gruntkowska, K.; and Richtárik, P. 2025. Gluon: Making Muon & Scion Great Again! (Bridging Theory and Practice of LMO-based Optimizers for LLMs). arXiv:2505.13416.

Richtárik, P.; Sokolov, I.; and Fatkhullin, I. 2021. EF21: A New, Simpler, Theoretically Better, and Practically Faster Error Feedback. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 4384–4396.

Safaryan, M.; and Richtárik, P. 2021. Stochastic Sign Descent Methods: New Algorithms and Better Theory. In *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, 9224–9234. PMLR.

Seide, F.; Fu, H.; Droppo, J.; Li, G.; and Yu, D. 2014. 1-bit stochastic gradient descent and its application to data-parallel distributed training of speech DNNs. In *Interspeech 2014*, 1058–1062.

Shah, I.; Polloreno, A. M.; Stratos, K.; Monk, P.; Chaluvaraju, A.; Hojel, A.; Ma, A.; Thomas, A.; Tanwer, A.; Shah, D. J.; Nguyen, K.; Smith, K.; Callahan, M.; Pust, M.; Parmar, M.; Rushton, P.; Mazarakis, P.; Kapila, R.; Srivastava, S.; Singla, S.; Romanski, T.; Vanjani, Y.; and Vaswani, A. 2025. Practical Efficiency of Muon for Pretraining. arXiv:2505.02222.

Sun, T.; Wang, Q.; Li, D.; and Wang, B. 2023. Momentum Ensures Convergence of SIGNSGD under Weaker Assumptions. In *Proceedings of the 40th International Conference on Machine Learning*, volume 202 of *Proceedings of Machine Learning Research*, 33077–33099. PMLR.

Takezawa, Y.; Koloskova, A.; Jiang, X.; and Stich, S. U. 2026. FedMuon: Federated Learning with Bias-corrected LMO-based Optimization. In *International Conference on Learning Representations (ICLR)*. arXiv:2509.26337.

Tang, H.; Yu, C.; Lian, X.; Zhang, T.; and Liu, J. 2019. DoubleSqueeze: Parallel Stochastic Gradient Descent with Double-Pass Error-Compensated Compression. In *Proceedings of the 36th International Conference on Machine Learning*, volume 97 of *Proceedings of Machine Learning Research*, 6155–6165. PMLR.

Thérien, B.; Huang, X.; Defazio, A.; Rish, I.; and Belilovsky, E. 2025. MuLoCo: Muon is a practical inner optimizer for DiLoCo. arXiv:2505.23725.

Yang, G.; Hu, E. J.; Babuschkin, I.; Sidor, S.; Liu, X.; Farhi, D.; Ryder, N.; Pachocki, J.; Chen, W.; and Gao, J. 2021. Tuning Large Neural Networks via Zero-Shot Hyperparameter Transfer. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 34, 17084–17097. arXiv:2203.03466.

Yang, G.; Simon, J. B.; and Bernstein, J. 2023. A Spectral Condition for Feature Learning. arXiv:2310.17813.

Zhang, X.; and Gao, H. 2025. On Provable Benefits of Muon in Federated Learning. arXiv:2510.03866.